PointBase Embedded PointBase Server

Multi-tiered applications like application servers, web servers, server-based E-commerce, sales force automation, customer relationship management, and marketing automation require a persistent data store capable of synchronizing with enterprise databases. A Java™ database in a workgroup environment is an ideal data store for these applications. Utilizing PointBase UniSync, PointBase Server and PointBase Embedded provide a seamless storage and synchronization solution for extending the enterprise information across multi-tiered networks of servers, desktops, laptops and smart devices.

PointBase Server-for networked applications

PointBase Sever provides robust relational data management for traditional client-server networked applications built on Internet standards for providing ubiquitous data access from anywhere on the Net. PointBase Server allows customers, partners, and employees around the world to access, share, and synchronize data through standard Internet tools and interfaces such as; HTTP, Java, and browsers.

PointBase Server provides a reliable and scalable database for e-commerce, application servers, web servers and other internet-based applications. The server design includes a multi-threaded architecture to ensure transactional integrity and performance under high transactional loads, row-level locking to minimize data contention and maximize performance, SQL security and encryption, multi-user network JDBC connections and transaction management.

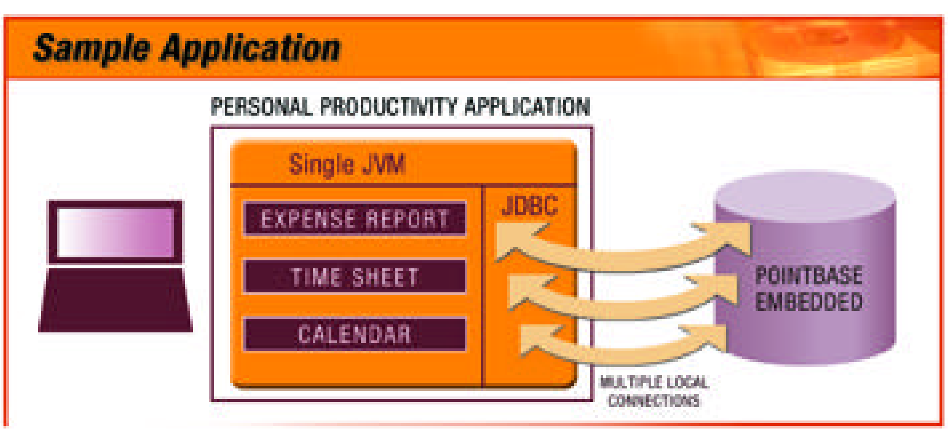
PointBase Embedded-for embedded applications

Integrated Java applications provide the users the ability to manage multiple types of information all at once in an integrated environment. These applications require the ability to persist the information locally and later synchronize with enterprise databases. Since the application handles different types of related information at once, it requires multiple concurrent connections to the local data store. Examples of these new applications include network management, sales force automation, marketing automation, and productivity applications. The diagram below depicts a Personal Productivity application running on a laptop. The application needs an embedded database that is based upon standards, is self-administering, has a low cost of ownership, allows multiple local connections, and is platform independent.

PointBase Embedded is a multiple connection database designed specifically for embedding inside Java applications and Internet appliances. The multi-threaded architecture allows multiple connections from within the same JVM. It provides full relational database functionality, small footprint, extensibility, synchronization with enterprise databases, near zero administration and a very low cost of ownership.

A transparent RDBMS in a jar-complete embedded functionality

The RDBMS ships as a Jar file which can be very easily packaged with an application. The end-user does not have to install the RDBMS separately. The application developer can configure the RDBMS based upon the application's requirements, making the underlying database transparent to the end-user, and providing complete embedded functionality.



PointBase Embedded | PointBase Server

100% Pure Java™
- Certified by KeyLabs
- Works on all platforms with a Java virtual machine.

Near-Zero Administration
- Automatic recovery
- Run-time database statistics
- Cost-based query optimizer
- Shutdown command

Security
- SQL security for assigning privileges and roles
- Complete data encryption (database and network levels)
- Standard encryption algorithms provide 40-128 bit encryption

Dynamic Footprint Architecture
- Object-oriented design facilitates flexible, modular implementation of applications
- Modularity allows applications to include only the required features for deployment

Transparent Data Management
- Fully integrated installation
- Minimal system resource requirements
- 100% Pure Java cross-platform support
- Scalable architecture accommodates application growth

Standard Application Development Tools and Interfaces
- WebGain (Symantec) VisualCafé, Sun Forte for Java and Motorola (Metrowerks) CodeWarrior bundle PointBase. Being used with major third party tools including Microsoft Visual J++, IBM VisualAge, and Inprise JBuilder
- SQL-92 Entry and Transition Levels
- SQL-99 (some)
- JDBC 1.2, and some JDBC 2.0
- Support for IBM DB2, Oracle SQL, Sybase, and MS SQL statement sets

DDL
- alter table add/drop constraints & columns
- connect/disconnect
- create/drop function
- create/drop procedure
- create/drop index
- create/drop schema
- create/drop table
- create/drop user roles
- grant/revoke privileges
- includes: default values, null, primary key, foreign key, check constraints, full referential integrity
- variable page sizes for individual tables and indexes, as well as blob/clob columns
- identity columns

DML
- insert
- update
- delete
- SQL functions in values clause, set clause, and predicates
- Scrollable, insensitive, updateable cursors
- JDBC 2.0 support for batch operations

DQL
- multi-table joins
- cost-based query optimizer
- on-line query statistics
- predicates: comparison, in, like, between
- SQL functions in select list and predicates
- aggregates (distinct, count, sum, max, min, avg, abs)

DCL
- call
- return
- set path
- set assignment
- signal
- switchlogfile
- values
- unload
- load

Scalar functions
- add
- cast
- char_length
- concatenation
- current_date
- current_schema
- current_time
- current_timestamp
- current_user
- divide
- extract
- lower
- multiply
- octet_length
- position
- substring
- subtract
- trim
- upper

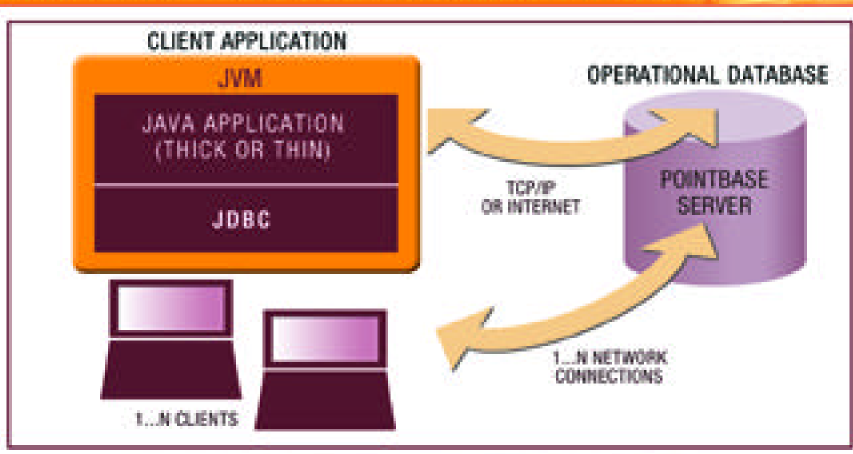
Transactions
- start transaction
- set transaction
- set datalog
- savepoint
- rollback to savepoint
- rollback
- commit
- isolation levels: read_uncommitted, read_committed, repeatable_read, serializable

Internationalization
- Unicode (international character set) support at the database, table and schema levels
- Locale settings (country and language)

Data Types
- blob
- BigInt
- char
- clob
- date
- decimal
- double
- float
- integer
- numeric
- real
- smallint
- time
- timestamp
- varchar

Backup
- Online Database backup
- API to load/unload Database

A Typical Client/Server Application Scenario



PointBase

PointBase Embedded

PointBase Server